NATIONAL TRANSPORTATION SAFETY BOARD
Office of Research and Engineering
Materials Laboratory Division
Washington, D.C. 20594

February 2, 2016

MATERIALS LABORATORY FACTUAL REPORT Report No. 16-006

A. ACCIDENT INFORMATION

Place	:	Lakeland, Florida
Date	:	April 20, 2015
Vehicle	:	Mooney M20TN
NTSB No.	:	ERA15FA191
Investigator	:	Shawn Etcher, AS-ERA

B. COMPONENTS EXAMINED

Throttle and mixture control assemblies.

C. DETAILS OF THE EXAMINATION

Overall views of the throttle and mixture control assemblies are shown in figure 1. Both assemblies were discolored and oxidized consistent with exposure to high heat. Remnants of aluminum alloy components on the assemblies were distorted, oxidized, flowed, and missing consistent with having been molten due to exposure to the heat.

The accident airplane reportedly had approximately 15 hours time in service since new at the time of the accident. According to a representative of Mooney International Corporation (Mooney) after referring to manufacturing records, the installed throttle control assembly was Orscheln Products, L.L.C. (Orscheln) part number 660226-027, and the installed mixture control assembly was McFarlane Aviation, Inc. (McFarlane) part number 660226-005. No identifying information was found on the throttle control assembly, but the overall construction was consistent with engineering drawings of the part provided by Orscheln. The manufacturer and part number for the mixture control matching the Mooney manufacturing records was found on the side of the swivel tube at the output end of the mixture control assembly.

During the examination of the throttle control assembly, representatives from Orscheln participated in the examination. A group examination of the throttle control assembly occurred on September 30, 2015.

1. Throttle Control Assembly

Closer views of the input and output ends of the throttle control assembly are shown in figure 2. The position of the input control had been documented on-scene with a line of white paint visible on the release rod and control tube in the upper image